ιTERIALS FOR LOW-FREQUENCY NOISE REDUCTION: A REVIEW

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ation vehicles is a growing
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re known for their disruptive
ι have harmful effects on the
residents. The continuous ex-
urban environments can lead
, and even more severe long-
g hypertension, stroke, and

ials such as foams exhibit fa-
racteristics. However, very
: required for absorbing low-
them unpractical in most ap-
: limited design tailorability,
employed in complex struc-

: innovative materials engi-
ol the propagation of sound
)le with traditional materials.
t of periodic arrangements of
interact with sound waves at
lesired acoustic effects [3].
:i) to contribute valuable in-
ıl and its practical implemen-
d (ii) to propose a metamate-
e manufacturing and capable
reduction in helicopters.

[aterials

ıovative approach used in the
to overcome the thickness re-
e low-frequency absorption
s. Instead of using thick ma-
volves creating intricate pat-
nd waves to traverse through
This enables the metasurface

micro-lattices. The effective thickness and intrinsic losses of each helicoidal cavity can be independently adjusted by varying their macro- and micro-structures. The results show almost perfect absorption over the frequency range 1000 – 2000 Hz that is not achievable with homogeneous porous materials or single helicoidal cavities.

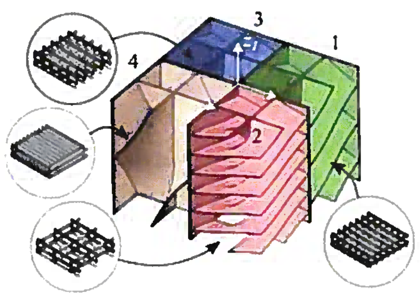

Figure 1. Schematic of parallel assembly of four folded metaporous materials [4].

2.2 Double Porosity Metamaterials

Double porosity metamaterial (DPM) refers to a type of porous material that has two levels of porosity: a meso-scale porosity and a micro-scale porosity.

Zhao *et al.* [5] designed a DPM, incorporating labyrinthine channels for meso-porosity and utilizing melamine foam to achieve microporosity. The DPM exhibits significantly lower frequency sound absorption compared to the homogenous porous materials with equal thickness. The distribution of sound pressure and velocity indicates that the improved sound absorption is brought about by two factors: the resonance of the labyrinthine channel and the hybrid resonance that occurs between the porous layer and the labyrinthine channel. These phenomena contribute to the overall enhancement in sound absorption. Moreover, the DPM offers greater design flexibility to customize the sound absorption spectrum in the desired frequency range.

2.3 Hybrid Acoustic Metamaterials

absorption capabilities, both in the low-frequency range and across a broader spectrum. The helical tubes would provide a smooth and continuous pathway for the sound waves to be trapped in and dissipated. Coiling is a highly effective method for compactly containing long tubes within limited volumes. In a subsequent work [7], the authors demonstrated that the helical conformation is the most efficient way to fold a tube to provide compact and structurally stable tubes. Furthermore, increasing the length of the tubes causes the resonance to shift towards lower frequencies, further optimizing the material's performance.

2.4 Acoustic Black Holes

An acoustic black hole (ABH) is a sound-absorbing structure designed to gradually slow down and weaken the sound waves as they travel towards the inner part of the structure. They are often comprised of a series of rings with different radii positioned along the inner wall of a rigid cylindrical tube, forming a conical shape. At the tip, there is an extremely small diameter, causing the sound waves to be reflected back and forth multiple times, resulting in a substantial reduction of sound energy and effective sound absorption [8].

Xiaoqi and Li [9] suggested combining an ABH sound absorber with microperforated panels (MPPs) as a liner covering the inner surface of the rings (Fig. 3). The designed structure demonstrates near perfect absorption performance in a rather low and broad frequency range from 200 Hz to 3000 Hz, with much fewer fluctuations of the absorption coefficient value compared to traditional structures.

MPPs offer high acoustic resistance and low acoustic reactance, which makes them a highly effective sound absorption material. The integration of the MPP with the cavity behind the panel creates a resonant system, facilitating the interaction of sound waves passing through the micro-perforations, thereby enhancing sound absorption within a specific frequency range. Furthermore, the MPP dissipates sound energy through viscous losses and frictional effects, contributing to the overall sound absorption performance of the ABH absorber. Therefore, incorporating the MPP into the ABH absorber significantly improves its sound absorption capabilities across both broadband and low-frequency ranges, making it a more versatile and efficient solution for noise control applications [9].

3 Conclusion

In conclusion, acoustic metamaterials offer innovative solu-

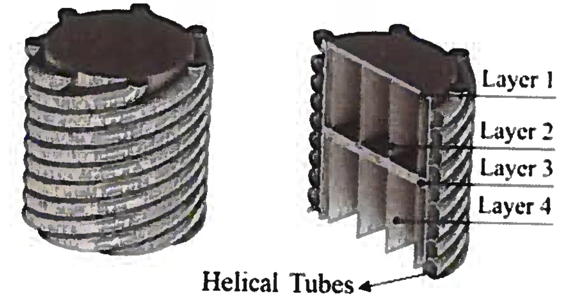

Figure 2. Schematics of hybrid metamaterials [7].

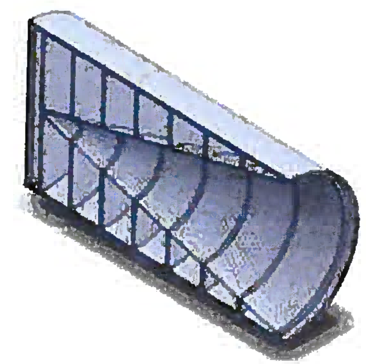

Figure 3. Schematic of acoustic black hole and microperforated panel as a liner [9].

Acknowledgments

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